Disjoint sets: exercises

21-E01 Partitions and repeated UNION calls

Insert ab, cd, bc, ab, ef, de on a..f. List every partition, successful-merge count, and component count. With all MAKE-SET calls first and one UNION per edge, each using two internal finds, what is external API count m? Which general invariant fails if same-root LINK still increases rank?

21-E02 The actual cost of naive lists

Create $x_1..x_6$, then perform UNION(x_2,x_1), UNION(x_3,x_2), …, UNION(x_6,x_5). For second-list-to-first appending and shorter-to-longer appending with first-set tie breaking, list rewrites per merge, final representative, and total rewrites. Prove the quadratic naive cost and linear weighted cost for this sequence. Why is weighted union not linear on every sequence?

21-E03 Per-member growth in an eight-element weighted list

Keep the first set on equal sizes and merge (a,b), (c,d), (e,f), (g,h), (a,c), (e,g), (a,e). Count each member's set rewrites, h's successive containing sizes, and total rewrites. For $n=2^k$, give the balanced-merge total and prove at most floor($\log_2 n$) rewrites per member.

21-E04 The same operations as a forest

Use rank with first-root tie breaking on the sequence from 21-E03. Give every final parent and rank, both passes of FIND(h), and the later FIND(d). If the second pass overwrites parent[x] before continuing with x=parent[x], which nodes are missed? Does returning the correct representative mean full compression was completed? May find reset rank to current depth?

21-E05 Rank and current height differ

Construct a legal rank-2 root of current height 1, listing the required unions and finds. Prove the 2^r membership bound and rank$\leq$floor($\log_2 n$). Explain why shrinking a nonroot's current subtree does not reduce its rank. For $n=10^6$, what is the maximum rank and minimum bit count for the rank value alone?

21-E06 Numerical levels and iteration counts

Take n=16, α=3, and a balanced rank tree with path h(rank 0)→g(rank 1)→e(rank 2)→p(rank 3)→r(rank 4), with self-parent root r. Compute the levels, iter counts, and phi of g,e,p before and after compression, and the path's change in Phi. Identify good nodes and exceptions. Under which conditions are level and iter nondecreasing? Derive $A_1(j)$, $A_2(j)$, $A_3(1)$, and distinguish composition from numerical exponentiation.

21-E07 Potential accounting needs fixed constants

A find visits s=7 nodes with α=3 and at least g=2 good nodes. Actual cost is at most 2s, and Ψ=3Φ. Bound its amortized cost and prove O(1+α) generally with fixed C>c. Four further operations have actual costs (3,7,2,8), while Ψ starts at zero and becomes (4,2,9,6). List amortized costs and verify the telescoping identity.

21-E08 Single calls, sequence cost, and small n

Compare single-find and sequence costs for rank alone, an arbitrary uncompressed chain, and rank with compression. Give a rank sequence reaching depth $\log_2 n$. For $f \geq 1$ repeated finds of one deepest leaf with no intervening UNION, compare totals with and without compression; count construction separately. List α(0),α(2),α(3),α(7),α(8),α(2047),α(2048). Why is the complete bound O(m(1+α))? At n=0, is any FIND input valid?

21-E09 Minimum-key queries and directed graphs

Why does always retaining the smaller-key root invalidate the rank theorem? Give an O(1) summary update that preserves rank linking and supports minimum-key queries. Use the one-edge directed graph a→b to distinguish union connectivity from SCC. Why can the original three DSU operations not maintain connectivity under edge deletion?

21-E10 Resetting iter and unsorted path levels

Equal ranks retain the first root. Building an interval in rounds means merging adjacent singletons, then adjacent equal-size blocks, until the interval's left endpoint is its root.
(1) Start with 32 singletons. Build 0..15, call FIND(12), build 16..31, then call UNION(16,0). List the path of another FIND(12), all path ranks, and each node's level, iter and potential before and after compression. Why does decreasing iter not increase potential?
(2) With 16 singletons, build 0..7, call FIND(2), build 8..15, and call UNION(8,0). List path ranks and levels before another FIND(2). Does this contradict monotonicity of one fixed node's level over time?
(3) Count actual good nodes G in each example and compare with $g = \max(0, s - (\alpha + 2))$. Does a path without good nodes imply an incorrect algorithm?

21-E11 A nonadjacent witness and a two-level jump

> Start with 128 singletons; equal-rank UNION retains the first root. Build 0..31 in rounds and call FIND(12). Build 32..63 and call UNION(32,0). Build 64..127 and call UNION(32,64). Each build merges adjacent singletons, then adjacent equal-sized blocks. Fix $n = 128, \alpha = 3$ throughout. Only now call FIND(14).

1. Give the path, ranks, parent ranks, and before/after level,iter,potential. Find the path and whole-forest potential changes without assuming omitted nodes have zero potential.
2. Identify every good node and its witness. Which node would an adjacent-level-only test miss? Compare actual G with lower bound g.
3. Check all five good-node inequalities for node 14. Must level increase by exactly one, or potential fall by exactly one?

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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https://github.com/xinhi6716/computer-science-self-study